Aeronautical System A Assignment

BY

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Aircraft angular velocity in Euler Angle equations derivation

The Airframe Angular velocity can be expressed by both airframe axes and Euler angle.
$[x, y, z]^T$: Airframe Fixed Coordinate System

The direction cosine matrix:

$$\mathrm{E}(\Phi, \Theta, \Psi) = \mathrm{E_x}(\Phi)\mathrm{E_y}(\Theta)\mathrm{E_z}(\Psi) \qquad (\mathrm{A1}-6)$$

$$\mathrm{E}(\Phi, \Theta, \Psi) = \begin{bmatrix} \cos\Theta\cos\Psi & \cos\Theta\sin\Psi & -\sin\Theta \\ \sin\Phi\sin\Theta\cos\Psi - \cos\Phi\sin\Psi & \sin\Phi\sin\Theta\sin\Psi + \cos\Phi\cos\Psi & \sin\Phi\cos\Theta \\ \cos\Phi\sin\Theta\cos\Psi + \sin\Phi\sin\Psi & \cos\Phi\sin\Theta\sin\Psi - \sin\Phi\cos\Psi & \cos\Phi\cos\Theta \end{bmatrix} (\mathrm{A1}-7)$$

$$\mathrm{E}^{-1} = \begin{bmatrix} \cos\Theta\cos\Psi & \sin\Phi\sin\Theta\cos\Psi - \cos\Phi\sin\Psi & \cos\Phi\sin\Theta\cos\Psi + \sin\Phi\sin\Psi \\ \cos\Theta\sin\Psi & \sin\Phi\sin\Theta\sin\Psi + \cos\Phi\cos\Psi & \mathrm{s}\cos\Phi\sin\Theta\sin\Psi - \sin\Phi\cos\Psi \\ -\sin\Theta & \sin\Phi\cos\Theta & \cos\Phi\cos\Theta \end{bmatrix} \quad (\mathrm{A1}-50)$$

$$\mathrm{E}^{-1} = \mathrm{E}^{\mathrm{T}} \qquad (\mathrm{A1}-51)$$

E is the Unity Matix.

Airframe Angular velocity can be expressed as:

$$\boldsymbol{\omega} = \dot{\Theta}\mathbf{j}_2 + \dot{\Psi}\mathbf{k}_{\mathrm{E}} + \dot{\Theta}\mathbf{i} \qquad (\mathrm{A1}-52)$$

Airframe angular velocity expressed in terms of airframe axes is:

$$\boldsymbol{\omega} = \dot{\mathrm{P}}\mathbf{i} + \dot{\mathrm{Q}}\mathbf{j} + \dot{\mathrm{R}}\mathbf{k} \qquad (\mathrm{A1}-53)$$

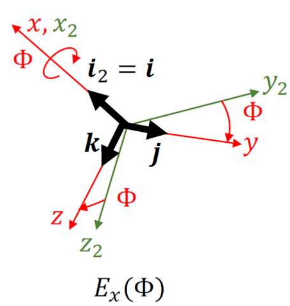

Fig.1 Rotation on aircraft x-axis

Determining the fixed-body axis component of $\mathbf{j}_2$:

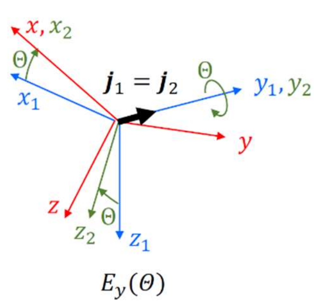

Fig.2 Rotation on aircraft y-axis

$$0 \times \mathbf{i}_2 + 1 \times \mathbf{j}_2 + 0 \times \mathbf{k}_2 = \mathrm{a_{j2}}\mathbf{i} + \mathrm{b_{j2}}\mathbf{j} + \mathrm{c_{j2}}\mathbf{k} \qquad (\mathrm{A1}-54)$$

$$E_x(\Phi)\begin{bmatrix}0\\1\\0\end{bmatrix}=\begin{bmatrix}a_{j2}\\b_{j2}\\c_{j2}\end{bmatrix} \qquad (A1-55)$$

$$E_x(\Phi)=\begin{bmatrix}1 & 0 & 0\\0 & cos\Phi & sin\Phi\\0 & -sin\Phi & cos\Phi\end{bmatrix}$$

$$a_{j2}=0$$
$$\cos\Phi=b_{j2}$$
$$-\sin\Phi=c_{j2}$$
$$\mathbf{j}_2=\mathbf{j}\cos\Phi-\mathbf{k}\sin\Phi \qquad (A1-56)$$

Determining the fixed-body axis component of $\mathbf{k}_E$:

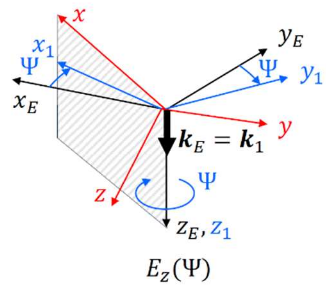

Fig.3 Rotation on aircraft z-axis

$$0\times\mathbf{i}_E+0\times\mathbf{j}_E+1\times\mathbf{k}_E=a_{kE}\mathbf{i}+b_{kE}\mathbf{j}+c_{kE}\mathbf{k} \qquad (A1-57)$$

$$E_x(\Phi,\Theta,\Psi)\begin{bmatrix}0\\0\\1\end{bmatrix}=\begin{bmatrix}a_{kE}\\b_{kE}\\c_{kE}\end{bmatrix} \qquad (A1-58)$$

$$a_{kE}=-\sin\Theta$$
$$b_{kE}=\sin\Phi\cos\Theta$$
$$c_{kE}=\cos\Phi\cos\Theta$$
$$\mathbf{k}_E=-\mathbf{i}\sin\Theta+\sin\Phi\cos\Theta\mathbf{j}+\cos\Phi\cos\Theta\mathbf{k} \qquad (A1-59)$$

By substitute (A1-56), (A1-59) into (A1-52), the angular velocity becomes:

$$\boldsymbol{\omega}=\dot{\Theta}(\mathbf{j}\cos\Phi-\mathbf{k}\sin\Phi)+\dot{\Psi}(-\mathbf{i}\sin\Theta+\sin\Phi\cos\Theta\mathbf{j}+\cos\Phi\cos\Theta\mathbf{k})+\dot{\Theta}\mathbf{i}$$

$$\boldsymbol{\omega}=(\dot{\Phi}-\dot{\Psi}\sin\Theta)\mathbf{i}+(\dot{\Psi}\cos\Phi+\dot{\Psi}\sin\Phi\cos\Theta)\mathbf{j}+\left(-\dot{\Theta}\sin\Phi+\dot{\Psi}\cos\Phi\cos\Theta\right)\mathbf{k} \qquad (A1-60)$$

By equating with (A1-53):

$$P=\dot{\Phi}-\dot{\Psi}\sin\Theta \qquad (A1-61a)$$
$$Q=\dot{\Theta}\cos c+\dot{\Psi}\sin\Phi\cos\Theta \qquad (A1-61b)$$
$$R=-\dot{\Theta}\sin\Phi+\dot{\Psi}\cos\Phi\cos\Theta \qquad (A1-61c)$$

By solving equation (A1-61), aircraft angular velocity expressed in Euler angles become:

$$\dot{\Phi}=P+Q\sin\Phi\tan\Theta+R\cos\Phi\tan\Theta \qquad (A1-62a)$$

$$\dot{\Theta}=Q\cos\Phi-R\sin\Phi \qquad (A1-62b)$$

$$\dot{\Psi}=Q\sin\Phi\sec\Theta+R\cos\Phi\sec\Theta \qquad (A1-62c)$$

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